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Research Article

Data-Driven Machine Learning Model for Battery Life Prediction Across Electrode Materials

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Accurately predicting the remaining lifespan of lithium-ion batteries (LIBs) is crucial for manufacturing processes and safe, reliable usage. Battery lifespan prediction continues to face major challenges due to varying degradation processes, fluctuating operating conditions, and differences in electrode materials. Here, we combine commercial battery data charged and discharged under different electrodes and temperature conditions to build a data-driven machine learning model for cycle life prediction. The datasets include three types of commercial cathodes: $LiFePO_4$ (LFP), $LiNi_{0.86}Co_{0.11}Al_{0.03}O_2$ (NCA), and $LiNi_{0.83}Co_{0.11}Mn_{0.07}O_2$ (NCM), which were cycled under various conditions and temperatures. The charging and discharging dataset under a single cathode material, trained using the Elastic Net model, shows that the root mean square error (RMSE) reaches over 1528 cycles under different electrodes. Furthermore, our findings reveal that temperature plays a critical role in predictive accuracy, emphasizing the importance of incorporating cycling conditions into prediction models. With both cathode diversity and temperature effects considered during model training, all RMSE values dropped below 200 cycles. Notably, the mean absolute percentage error (MAPE) for NCA decreased from 64% to 27%. These outcomes highlight a promising approach for developing robust machine learning models capable of accurate battery performance prediction across varied conditions, contributing to safer and more reliable battery technology.

Keywords: cathode material; cycling condition; li-ion battery; machine learning

1. Introduction

In recent years, the importance of lithium-ion batteries (LIBs) has grown significantly across various fields, including electronic devices, electric vehicles, and grid energy storage, due to their high energy density, long cycle life, and cost-effectiveness [1–4]. Global electric car sales exceeded 40 million in 2023 and are projected to surpass 74.5 million by 2035, driven by carbon targets and growing consumer demand [5–7]. However, despite this rapid growth, the unpredictability of battery degradation remains a major challenge in their development and practical application. LIBs experience various degradation phenomena over time, leading to capacity and power fade and eventually causing battery failure. The degradation of battery cells is caused by irreversible chemical and physical processes, including the depletion of lithium inventory and the deterioration of active materials [8–11]. The rate and nature of this degradation are significantly influenced by the materials used in the electrodes, the patterns of battery usage, and the operating environments, such as temperature and the speed of charging and discharging [12–14]. As a result, the level of degradation differs between individual cells, making it challenging to ensure the consistent and reliable operation of LIBs. Therefore, it is essential for the battery management system to accurately predict battery performance to maintain safety and reliability [15, 16].

Battery performance prediction models are generally divided into two main types: physics-based models and data-driven models [17]. Physics-based models [18–23] approach battery performance prediction by simulating electrochemical

processes within the battery through mathematical and physical frameworks. These models utilize partial differential equations to describe Li-ion movement, electrical potential distribution, and reaction dynamics, offering a detailed understanding of battery operation and degradation processes. For example, Baghdadi et al. [18] presented a linear relationship between remaining capacity and voltage after a 30 min rest when the cell is fully charged, and the percentage error is less than 2% in capacity estimation. Tang et al. [19] proposed the model to predict the state of health (SoH) using incremental capacity (IC) analysis, a method used for the analysis of aging mechanisms, with a coefficient of determination (R^2) are greater than 0.9 when the regional voltage in the neighborhood of peak voltage was 200 mV. However, physics–based models often require frequent parameter updates for each cell across cycles, which significantly increases the computational and time costs [24].

On the other hand, data-driven approaches [25–31] utilize machine learning techniques, various transformations, and statistical methods to extract key features from raw data. These methods are highly effective at identifying nonlinear relationships and are widely adopted due to their efficiency and accuracy [32]. Machine learning techniques are extensively applied to predict various aspects of battery performance, such as SoH, end-of-life, and future discharge capacity. For instance, Severson et al. [25] proposed a model that predicts the cycle life of commercial $LiFePO_4$ (LFP) || graphite cells using only data from the first 100 cycles, achieving a test error of 9.1%. Zhu et al. [26] constructed the machine learning model using variation, skewness, and maxima from voltage relaxation after fully charging as input features to predict the discharge capacity, achieving a test root mean square error (RMSE) of 1.1% with the XGBoost method for three commercial batteries. Also, Sahinoglu et al. [27] proposed Gaussian process regression (GPR) to estimate the state of charge (SoC) based on voltage, current and temperature, showing 0.8% error in the estimation. Recent studies have further adopted advanced architecture based on deep learning, such as transformers and multimodal models, to further improve battery performance prediction [33–35].

However, most existing research focuses on specific cathode materials tested under restricted conditions. Although these models excel at capturing complex nonlinear relationships in battery degradation, their effectiveness heavily depends on the availability of diverse and high-quality training datasets [24, 36]. As a result, their predictions may lack accuracy or physical significance, particularly when applied to out-of-sample data that deviate from the patterns used during training. This dependence on electrode materials and cycle conditions restricts the versatility of the models, making it difficult to extend their application to other materials or operating environments. To address these limitations, it is essential to use diverse and comprehensive training data that can capture the complex and various degradation mechanisms across different battery materials and cycling conditions. This approach enables the exploration of complex degradation behaviors across different materials and usage scenarios, ultimately improving prediction accuracy and adaptability.

In this study, we aim to overcome the limitations of lifespan prediction accuracy by constructing a dataset that includes various cathode materials and cycling conditions. Specifically, we incorporate data from $LiFePO_4$ (LFP), $LiNi_{0.86}Co_{0.11}Al_{0.03}O_2$ (NCA), and $LiNi_{0.83}Co_{0.11}Mn_{0.07}O_2$ (NCM) batteries, each cycled under various conditions, to build and evaluate a more robust and generalized model. We first investigate how variations in cathode materials and cycling conditions affect the input feature used for prediction. Then, we work to improve the predictive power of models by using diverse battery data. Increasing the diversity in training data is crucial for machine learning models to generalize better and make robust predictions under real-world conditions, where battery performance can vary significantly based on material properties and cycling conditions. Our approach leverages this diversity in training data to capture a wider spectrum of degradation patterns, ultimately enhancing the model's applicability across different battery systems.

2. Method

To evaluate the robustness of a feature, we utilize the battery datasets cycled under full charge and discharge conditions, which are commonly used for battery performance prediction. LFP, NCA, and NCM battery data were sourced from publicly available databases. The Massachusetts Institute of Technology (MIT) [25] database includes 124 commercial LFP batteries, divided into three batches. The Zenodo [26] database provides data for three commercial 18,650 battery types: NCA, NCM, and NCA + NCM. In this work, data from the NCA + NCM batteries were excluded due to variability in charge and discharge conditions between the 10th and 100th cycles, as well as anomalous time intervals during discharge, making accurate feature generation challenging. Thus, the analysis focuses on data from three commercial cathode materials: LFP, NCA, and NCM.

All discharge curves were fitted using spline fitting. To standardize the feature generation method, it was essential to ensure a consistent voltage range within each cathode material. Data from LFP batteries don't have a relaxation stage after being fully charged, resulting in a uniform voltage range of 2.5 to 3.7 V. However, data from NCA and NCM batteries include a 30 min relaxation stage, which causes their voltage ranges to vary due to relaxation-induced voltage drop. To ensure consistency, we compared the starting voltage values of each discharge curve and took the lowest voltage among them as the reference. Then, all voltage data points above this reference point were removed. As a result, NCA batteries cover a voltage range of 2.65 to 4.0 V, and NCM batteries cover a voltage range of 2.5 to 3.95 V. This ensured that the difference between the 10th and 100th discharge curves was calculated over the standardized voltage ranges for each cathode material.

Additionally, the log transformation used during feature generation served to adjust the data distribution, making it closer to a normal distribution and reducing skewness [37]. This transformation minimized the impact of outliers and contributed to more balanced model training. Afterward, standard normalization was conducted using scaling parameters derived

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Table 1: Basic information and cycling conditions for five datasets.

Dataset	Electrode active material	Voltage range (V)	Nominal capacity (mAh)	Cycling temperature (°C)	Number of cells	Data source
Dataset1	NCA/graphite	2.65–4.2	3500	25	21	TJU
Dataset2				45	26	
Dataset3	NCM/graphite	2.5–4.2	3500	25	22	
Dataset4				45	28	
Dataset5	LFP/graphite	2.0–3.6	1100	30	119	MIT

Note: All cells are commercial 18,650 cylindrical batteries.

from LFP features. These same parameters were then applied to NCA and NCM features, ensuring that LFP, NCA, and NCM features were all normalized using a consistent scaling standard.

The total data was divided into training and test datasets using a strict cell-level group-wise split, ensuring that data from the same cell never appeared in both training and test datasets to prevent data leakage. The training dataset was used to determine the hyperparameters of machine learning models and train the model, while the test dataset was reserved for model evaluation. Due to the small training dataset, we employed repeated K-fold cross-validation to account for fold-to-fold variability when determining optimal parameters. This method repeats the K-fold cross-validation process multiple times and averages error results across all folds and repetitions, which contributes to reducing the effect of random data splits. Furthermore, we considered both the magnitude and variability of the error across folds, using MSE_avg $+ \sigma^2$ as the error metric, where MSE_avg is the average of mean squared error (MSE) across folds and σ^2 is the variance of MSE across those folds. This approach helps reduce sensitivity to specific data and build a more robust and stable model.

RMSE and mean absolute percentage error (MAPE) were used to evaluate model performance using the test dataset.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \overline{y}_i)^2}, \tag{1}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \overline{y}_i}{y_i}\right| \times 100, \tag{2}$$

where y_i is the measured cycle life, $\overline{y}_i$ is the predicted cycle life and n is the total number of samples in the test dataset. The analysis code used to train and evaluate the machine learning models in this study is available in the Data Availability section. The original cycling data can be obtained directly from the MIT [25] and Zenodo [26] repositories to reproduce the model training and evaluation.

3. Results and Discussion

3.1. Data Generation and Feature Extraction. For this study, we employ a total of 216 battery samples with different cathode materials and cycling conditions, sourced from publicly available datasets [25, 26]. To investigate the effects of cathode materials and cycling conditions on data-driven models, the data were divided into five datasets based on cathode material and cycling temperature. Table 1 summarizes the electrode materials and cycling conditions for each dataset. The cycling data in Datasets 1–4 were generated by Zhu et al [26]. at Tongji University (TJU). Datasets 1 and 2 contain data on identical NCA batteries cycled at 25 and 45°C, respectively. Similarly, Datasets 3 and 4 contain data on identical NCM batteries cycled at 25 and 45°C, respectively. Dataset 5 contains data on LFP batteries cycled at 30°C, generated by Severson et al. [25] at the MIT. These LFP batteries were cycled under 72 different fast-charging strategies with multiple stages and discharged at a 4 C rate. In contrast, the NCA and NCM batteries followed a CC-CV charging and discharging protocol (Figure 1). TJU defines cycle life as the cycle at which 71% of its initial capacity remains, while MIT defines it as the cycle at which 80% of its initial capacity remains. These differences in EOL thresholds arise from intrinsic material characteristics. The 71% SOH criterion applied to NCA and NCM reflects their pronounced capacity fade near the end of life, whereas the 80% SOH threshold for LFP corresponds to its more gradual degradation profile and widely accepted practical cutoff. Such material-specific definitions of EOL constitute an important aspect of the physical heterogeneity that our model is intended to capture.

As shown in Figure 2a-c, even within the same battery type produced by the same manufacturer, battery performance varies due to factors such as inherent aging, cycling conditions, and environmental influences. Regardless of cycling conditions, NCA and NCM batteries generally exhibit a cycle life within 1000 cycles, whereas LFP batteries typically achieve a longer cycle life than NCA and NCM batteries (Figure S1a). Following Severson et al. [25], early prediction of cycle life for LFP batteries can be achieved using only the discharge voltage curves from the 10th and 100th cycles. The ΔQ_{100-10} (V) represents the change in discharge voltage curves between the 10th and 100th cycles across the entire operational voltage range. To extract a meaningful and simple representation from this raw data, statistical transformation was applied. Specially, the log variance of ΔQ_{100-10} (V) (log (variance [ΔQ_{100-10} (V)]) effectively captures battery degradation and has strong predictive power for LFP batteries. However, it remains uncertain whether this feature holds similar predictive power for other cathode materials. To explore the impact of different cathode materials on this feature, we generated log (variance [ΔQ_{100-10} (V)]) for NCA and NCM batteries cycled under different conditions, following the same methodology as the original study.

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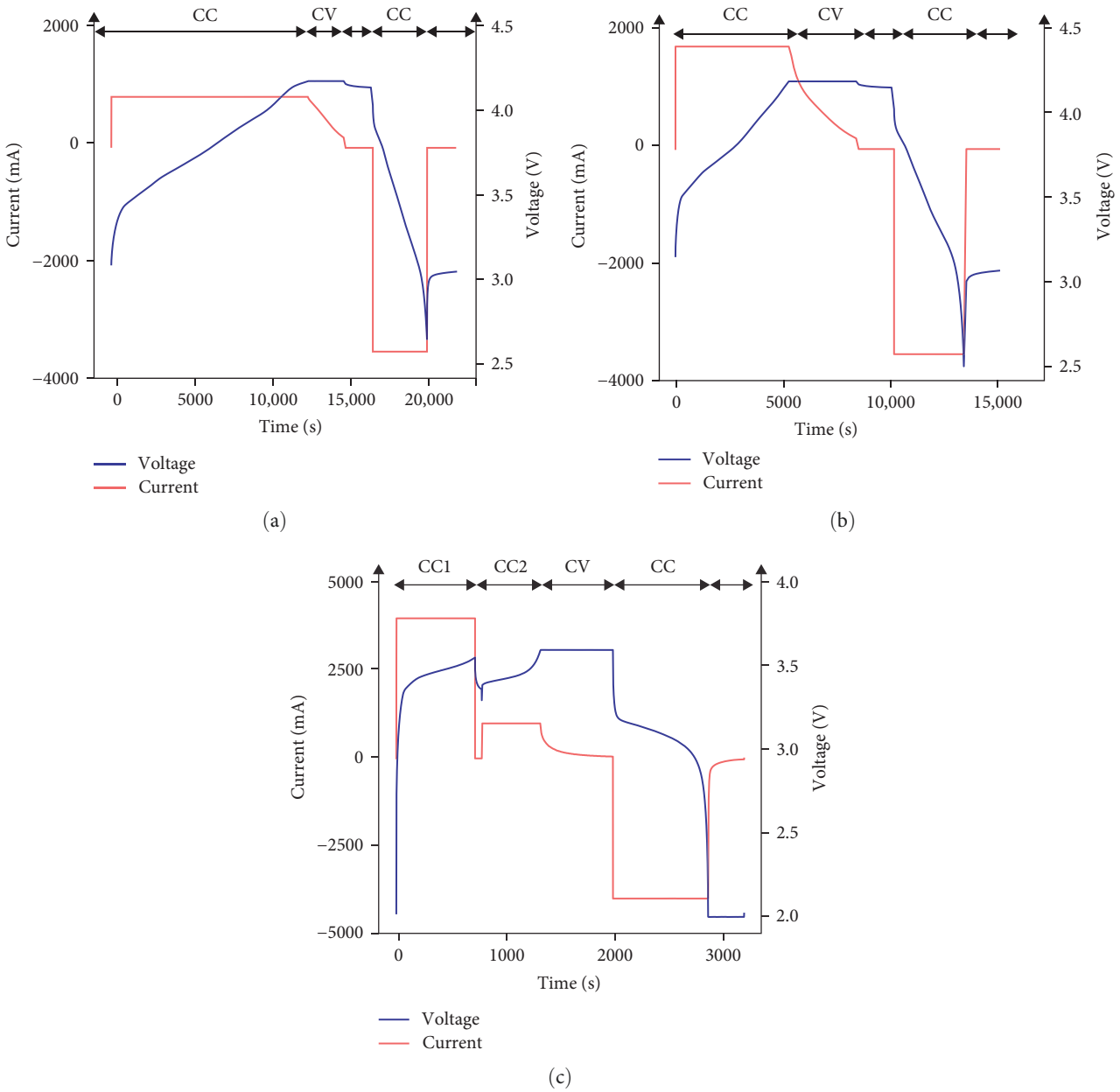


FIGURE 1: Voltage and current profiles during charge and discharge of three different battery cells. Each profile represents the first cycle of a battery: (a) NCA, (b) NCM, and (c) LFP. The current rate is defined according to the nominal capacity of each battery. The 1C rate is 3.5 A for both the NCA and NCM batteries and 1.1 A for the LFP batteries.

As illustrated in Figure 2d-f, the cathode material significantly affects the ΔQ_{100-10} (V) curve, resulting in distinct curve shapes for each material. Furthermore, the range of ΔQ_{100-10} (V) values also varies across materials, further emphasizing the influence of cathode material on the feature (Figure S1b).

To further analyze these changes, we calculated the correlation coefficient [38] to quantify the relationship between input feature and cycle life across different cathode materials (Figure S2). We examined three types of correlation: Pearson, Kernel, and Spearman (Table S1). Pearson correlation captures the linear relationship between data, whereas Kernel and Spearman correlations focus on nonlinear relationships in the data. When comparing these correlations across materials, both LFP and NCA batteries exhibit a negative correlation between the input feature and cycle life, while NCM batteries show a positive correlation. Although both LFP and NCA batteries exhibit a negative correlation coefficient, the strength of the correlation differs substantially. This reveals that the relationship between the input feature and cycle life varies significantly across cathode materials. Due to such diversity, it is challenging to define a single consistent trend that applies to all cathode materials. This inconsistency can be a barrier to building machine learning models, which train patterns across the training data.

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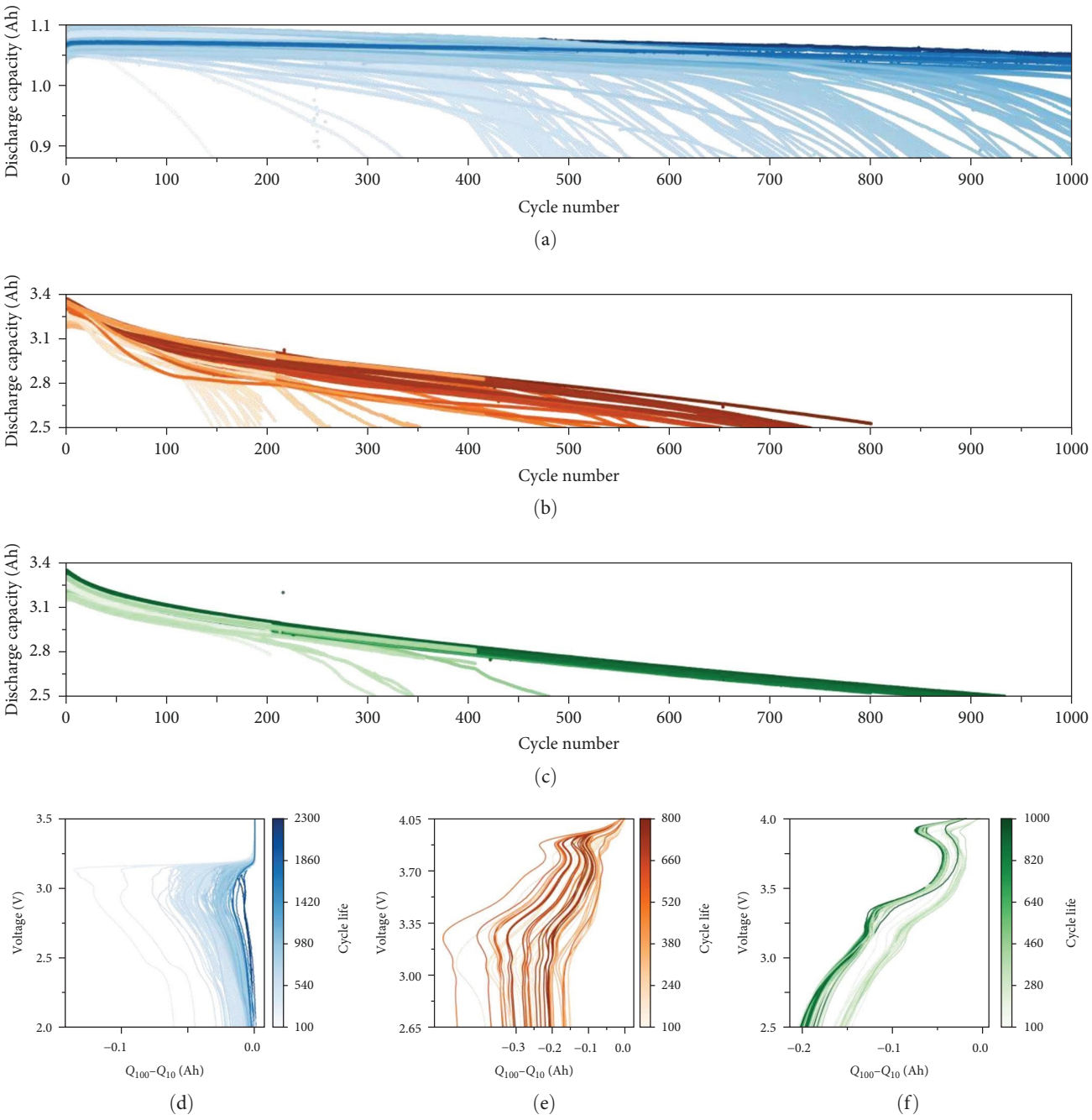


FIGURE 2: Capacity fade curves of different battery cathode materials and corresponding changes in ΔQ_{100-10} (V) curves. (a), (b), and (c) show the discharge capacity degradation over the cycle number for LFP, NCA, and NCM batteries, respectively, highlighting the distinct degradation patterns associated with each material. (d), (e), and (f) show the ΔQ_{100-10} (V) curves as a function of voltage for LFP, NCA, and NCM batteries. The changes in these curves reflect the unique degradation characteristics of each cathode material.

3.2. Influence of Data on Electrode Materials. Machine learning models have demonstrated higher efficiency compared to physics-based approaches; however, their accuracy depends on the relevance and representativeness of the extracted features. As shown in the previous correlation analysis, the relationship between the input feature and cycle life varies significantly across cathode materials, both in direction and magnitude. A strong correlation between input and output features is crucial for accurate and reliable prediction. For instance, log (variance [ΔQ_{100-10} (V)]) is highly correlated with the cycle life of LFP batteries, while it does not show the same strong correlation with NCA and NCM batteries. Given the inconsistent relationship between the feature and cycle life across cathode materials, we investigated whether this feature could provide reliable predictions across cathode materials using four different machine learning methods: Elastic Net [39], XGBoost [40], Support Vector Regression (SVR) [41], and GPR [42] (Table S2).

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All machine learning methods were trained on the same training dataset, and their accuracy was compared using RMSE as the error metric. Figure 3 illustrates the performance of models trained on each different cathode material. The results indicate that log (variance [ΔQ_{100-10} (V)]) has limited predictive power for NCA and NCM batteries, which is consistent with the weaker Pearson correlation coefficients of –0.34 for NCA and 0.65 for NCM. Furthermore, when the cathode material used in the prediction differed from the material used during training, the accuracy dropped significantly. These highlight the challenge of maintaining prediction accuracy across different cathode materials, due to the difficulty in capturing complex degradation mechanisms that vary with cathode material (Figure S3). These results show the dependency of predictive performance on the cathode material, suggesting that generalizing models across different cathode materials is challenging. To address this issue, we constructed a dataset with increased material diversity. We trained a model on a dataset that included data from all three different cathode materials, thereby enhancing the material diversity of the training data. XGBoost demonstrated consistently superior performance among the evaluated methods and was selected as the primary model due to its flexibility and tunable parameters.

We explored the effect of material diversity during model training by comparing the predictive power of three models trained on different datasets: one model trained solely with LFP data, a second model trained with both LFP and NCA data, and a final model incorporating data from LFP, NCA, and NCM batteries. In other words, the number of cathode materials included in the training dataset increased with each model. The workflow is illustrated in Figure 4. Due to differences in feature magnitudes, standard normalization was applied to the LFP features, and the same scaling was used for the NCA and NCM features. To ensure a leakage-safe evaluation, we adopted a strict cell-level group-wise split to the dataset, dividing it into training and test sets in a 7:3 ratio. This procedure ensures the data from the same battery cell never appear in both the training and test sets. This strategy was applied consistently across all evaluations, including the repeated K-fold cross-validation ($K = 3$, $N = 5$) used for hyperparameter tuning and model training. Using the identified hyperparameters, models were then trained on the entire training dataset and evaluated on the test dataset. Prediction accuracy was assessed using RMSE and MAPE as error metrics. RMSE represents the typical magnitude of errors, while MAPE indicates prediction accuracy relative to the true values as a percentage. Therefore, these error metrics can offer a balanced view by showing both absolute and relative prediction errors. To ensure more generalizable results, we repeated the process with three random splits of the dataset into training and test datasets. For each split, the entire process, including K-fold cross-validation, training, and evaluation, was repeated. This repetition helps reduce the influence of any single data split on model performance and provides a more robust estimate of generalization ability.

Figure 5 shows the changes in RMSE for each random split as the number of cathode materials used during training increases. The error values for NCA batteries were lower when the training dataset included NCA data compared to when it did not, and similarly for NCM. There was a significant reduction in MAPE, with nearly 50% for NCA and 34% for NCM (Table 2). Overall, error values decreased as the diversity of the training data increased, indicating that models trained with diverse materials generalize better across different materials (Figure S4 and Figure S5). Increasing the material diversity in training data allows models to better capture complex degradation patterns across different materials and improve overall predictive performance for various battery systems. Therefore, ensuring material diversity in training data is essential for robust prediction with machine learning models.

3.3. Influence of Temperature on Prediction Accuracy. In most battery performance predictions, it is essential to find features that accurately capture battery degradation, as they have a significant impact on the prediction accuracy. Various factors during cycling, such as temperature [43], depth of discharge (DOD) [44], and rate of charge and discharge [45], influence the battery degradation process, suggesting that using only transformed features may not sufficiently predict a battery's cycle life. Therefore, we further investigated the effect of cycling conditions on the feature. Among them, temperature is one of the influential factors that affect battery degradation [44]. For LFP batteries, capacity fade accelerates at higher temperatures [43, 44, 46], whereas for NCM batteries, capacity fade decreases as temperature increases [44, 47]. The relationship between temperature and battery degradation for NCA batteries is less clear, suggesting more complex degradation behavior [44, 47]. These variations indicate that each cathode material uniquely responds to temperature changes. As given in Figure 6, the ΔQ_{100-10} (V) curve clearly changes with temperature. Temperature conditions maintain the overall trend of the curve but alter the fine structure, shifting the location and intensity of the peak. This suggests that log (variance [ΔQ_{100-10} (V)]) is temperature-sensitive, indicating that the predictive accuracy can vary with cycling temperature. Given this, temperature should be considered when developing models to enhance accuracy. To account for thermal effects in the prediction, we developed a model using two input features: log (variance [ΔQ_{100-10} (V)]) and cycling temperature. The process of generating the training and test datasets, training the model, and evaluating its performance followed the same process as in the previous chapter. We compared the model accuracy when trained on only log (variance [ΔQ_{100-10} (V)]) with when trained on both log (variance [ΔQ_{100-10} (V)]) and cycling temperature, using the same training and test datasets. The results, shown in Table 3 and Figure 7, indicate that incorporating temperature significantly lowers RMSE and MAPE across all cathode materials, with RMSE values under 200 for each. In most cases, incorporating temperature as an additional feature leads to a reduction in prediction error across individual battery cells. This demonstrates that considering temperature as an input feature allows models to better capture thermal effects on battery degradation, thereby improving predictive accuracy.

Temperature conditions particularly enhance prediction accuracy for NCA, with both RMSE and MAPE values substantially reduced compared to other cathode materials, decreasing from 251 to 124 in RMSE and from 64% to 27%

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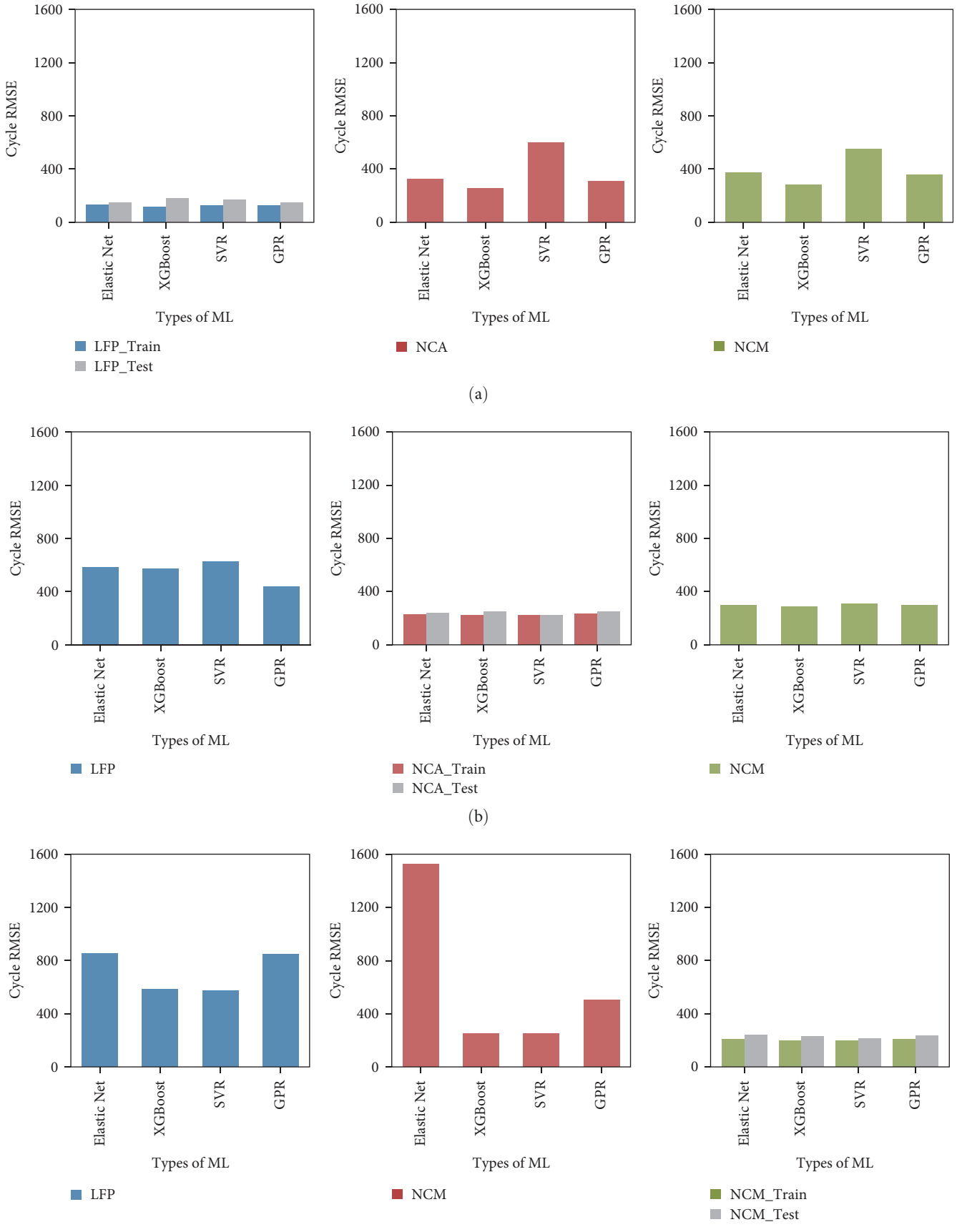


FIGURE 3: Poor predictive performance of models trained on data from a single cathode material. Results are shown for models trained only on (a) LFP, (b) NCA, and (c) NCM. Each machine learning method is trained on the same training dataset with similar RMSE levels, maintaining RMSE within ± 10 of the Elastic Net RMSE to ensure consistent performance. Despite using the same feature, the errors are higher for NCA and NCM than for LFP. Furthermore, models trained on data from a single material show poor generalizability, revealing the limited robustness across materials.

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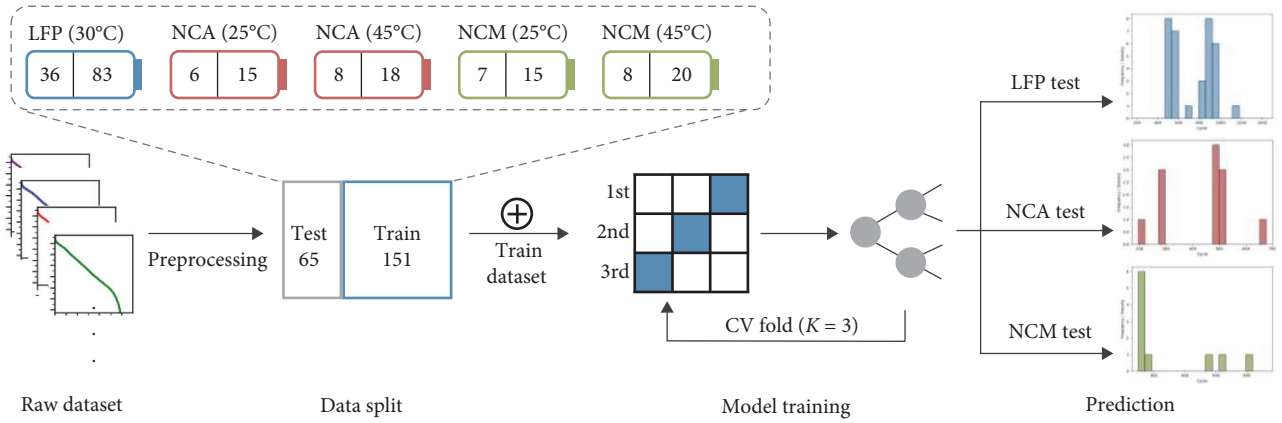


FIGURE 4: Workflow for developing a model incorporating data from LFP, NCA, and NCM batteries. First, log (variance $[\Delta Q_{100-10}$ (V]) is calculated from the 10th and 100th discharge curves. Second, each dataset is randomly split into training and test datasets with a 7:3 ratio to consider the effect of material and cycling conditions. Third, repeated cross-validation ($K = 3$, $N = 5$) is performed to optimize and determine the XGBoost model parameters, after which the model is trained on the entire training dataset. Finally, the model predicts the cycle life of each cell in the test dataset, and the test error is calculated to evaluate model performance.

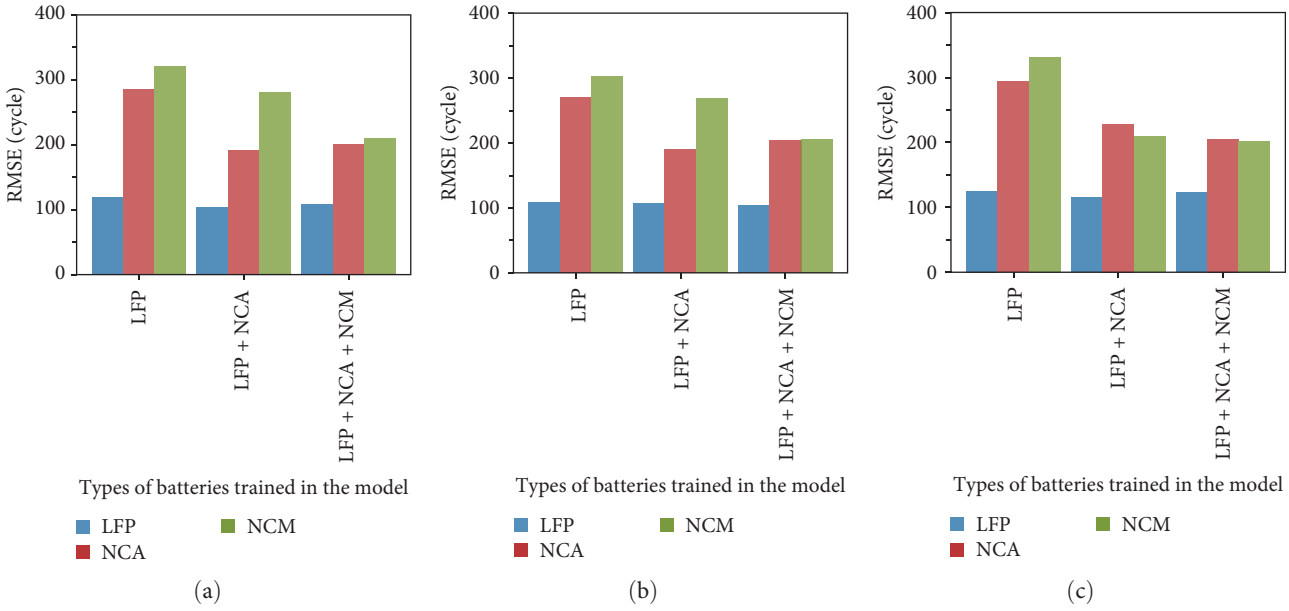


FIGURE 5: Test results for models trained on different combinations of cathode materials. Due to the small training dataset, we performed three random splits of the dataset to obtain more generalizable results. (a), (b), and (c) represent the results from three different dataset split. Each split was used to train model on different combinations of cathode materials: LFP, LFP + NCA, and LFP + NCA + NCM. Although differences in error values are observed among splits, including multiple cathode materials in the training dataset enhances prediction accuracy.

TABLE 2: Test errors of battery cycle life prediction for the model presented in Figure 5c.

Training set composition	RMSE (cycles)			MAPE (%)		
	LFP	NCA	NCM	LFP	NCA	NCM
LFP	122	270	303	13	102	120
LFP + NCA	104	190	268	11	52	103
LFP + NCA + NCM	104	203	206	11	49	68

Note: For models trained on only LFP data, cycle life predictions for LFP are made using only the LFP test dataset. For NCA and NCM batteries, which are not included in the training dataset, all available samples are used for evaluation.

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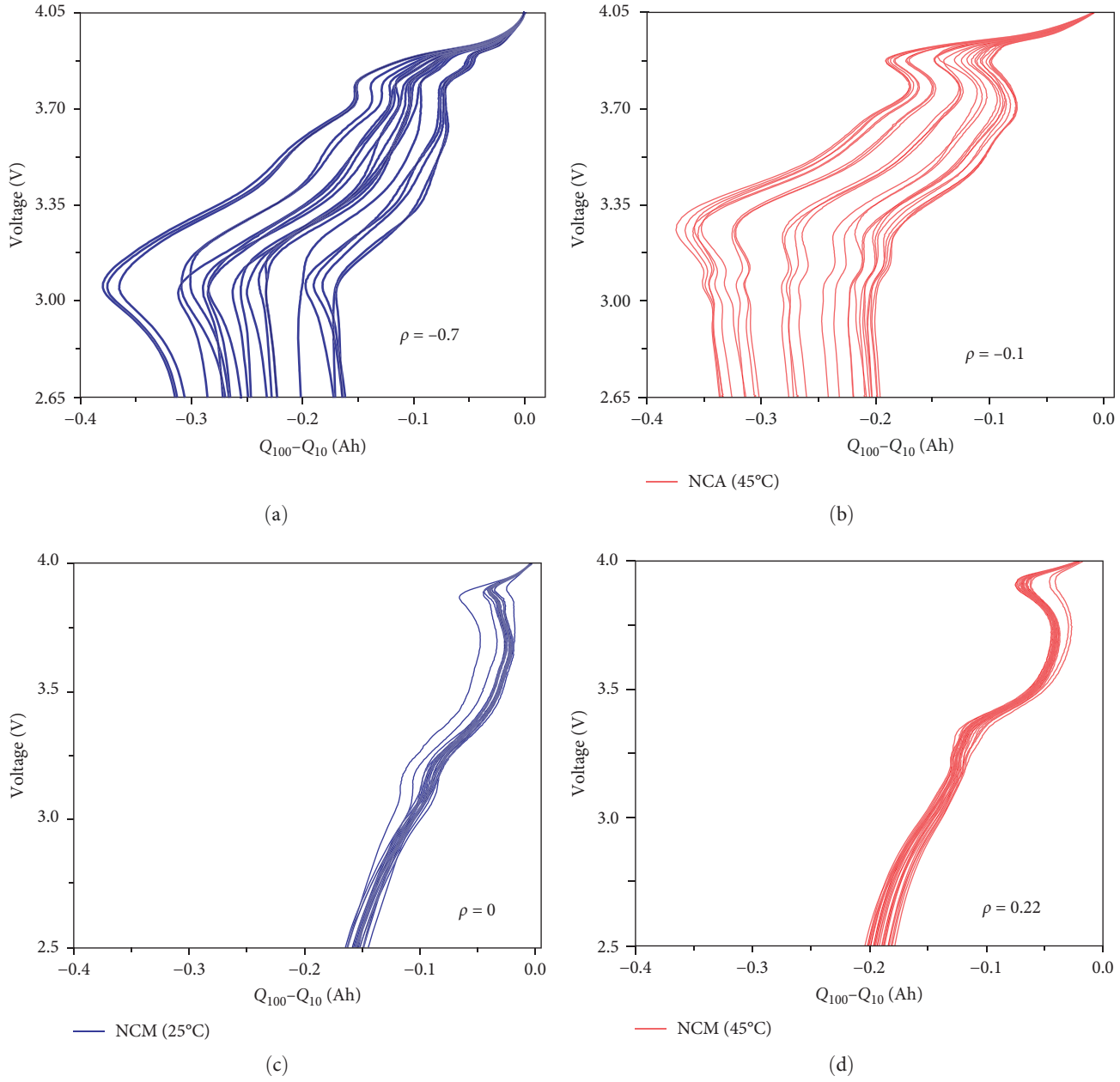


FIGURE 6: Temperature-dependent ΔQ_{100-10} (V) curves of NCA and NCM batteries and their corresponding correlation coefficients (ρ) with cycle life, as shown for (a) NCA at 25°C, (b) NCA at 45°C, (c) NCM at 25°C, and (d) NCM at 45°C. The overall trend remains consistent; however, fine structures, such as peak location and intensity, vary with temperature. These changes are particularly evident in the NCA feature curves, where at 45°C, the capacity difference between 10th and 100th cycle becomes more pronounced within specific voltage ranges, which also shift upward compared to those at 25°C. Additionally, the Pearson correlation, which was strong at 25°C ($\rho = -0.7$), becomes much weaker at 45°C ($\rho = -0.1$), indicating a change of predictive power.

in MAPE. This clear evidence of performance gain, achieved solely through the incorporation of material and condition diversity, validates our data-centric approach as an effective alternative to architectural improvements. Specifically, the significant reduction in MAPE (e.g., from 60% to 23% in NCA) demonstrates that leveraging data diversity is key to achieving a robust cross-material generalization capability that effectively addresses the heterogeneity of various battery chemistries and cycling conditions. This improvement corresponds to the significant variation in the ΔQ_{100-10} (V) curve for NCA as cycling temperature varies. Conversely, LFP remains unaffected by temperature. This can be attributed to two main factors. First, all LFP data were cycled at 30°C. Second, LFP batteries were cycled under fast-charging conditions, which diminish the fine structure of the ΔQ_{100-10} (V) curve. This phenomenon aligns with findings in IC analysis, where studies have shown that high charge and discharge rates lead to a simplification of the IC curve structure [48–50].

Additionally, while this study primarily focuses on the effect of temperature, other cycling conditions, such as charge

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TABLE 3: Comparison of test errors with and without temperature as an input feature. ΔQ_{100-10} represents the model using only log (variance [ΔQ_{100-10} (V)]) as the input feature, while ΔQ_{100-10} + TEMP model includes cycling temperature as an additional feature.

Input feature	RMSE (cycle)			MAPE (%)		
	LFP	NCA	NCM	LFP	NCA	NCM
$\Delta\ Q_{100-10}$	103	251	178	11	60	41
$\Delta\ Q_{100-10}$ + TEMP	102	124	169	11	23	38

Note: Incorporating temperature significantly reduces test error for NCA and NCM batteries.

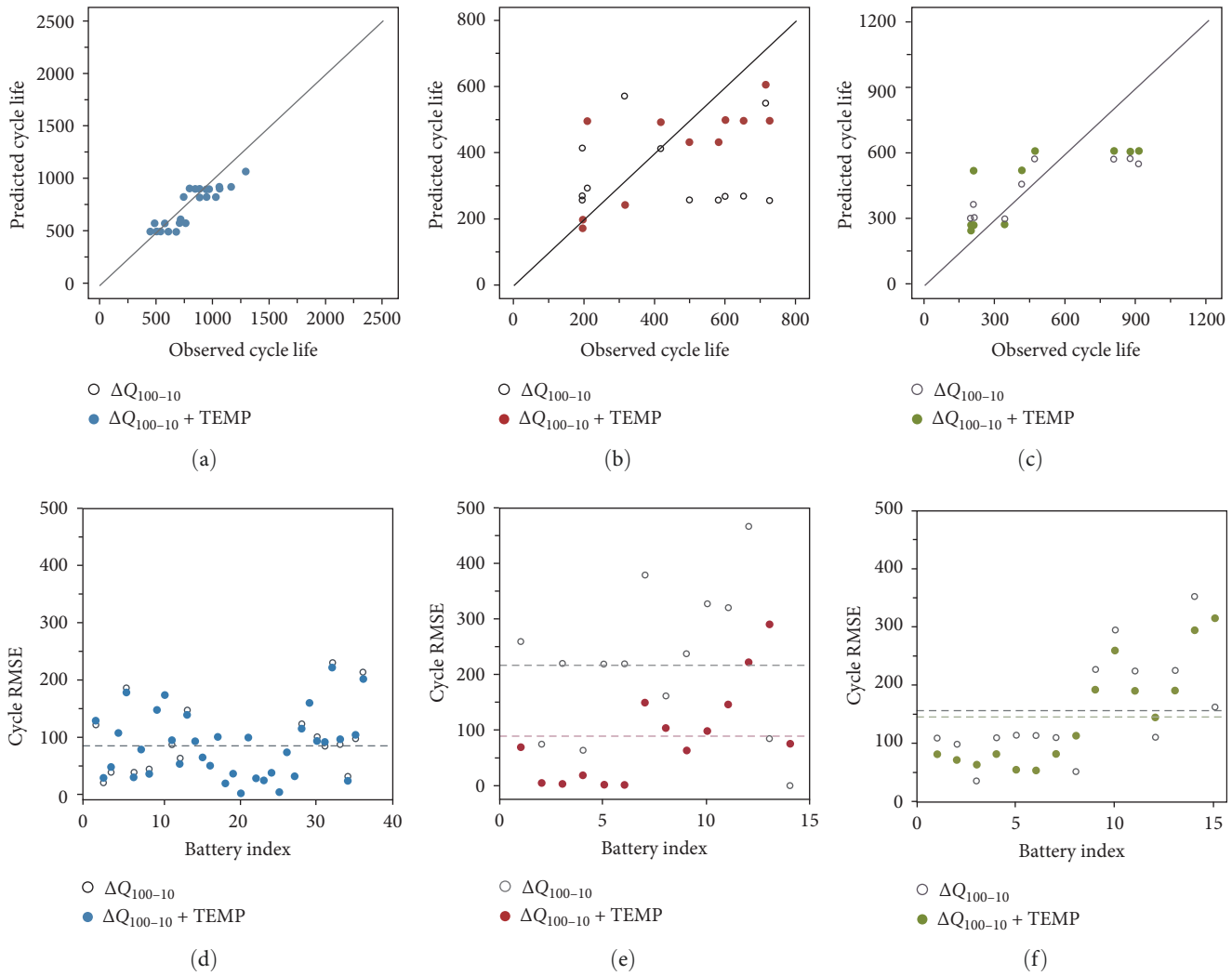


FIGURE 7: Results of models incorporating material diversity and temperature in the training process. (a), (b), and (c) show the predicted cycle life vs. observed cycle life for LFP, NCA, and NCM batteries. Two models are compared: one considering only material diversity and another considering both diversity and temperature. (d), (e), and (f) present the corresponding RMSE of all battery cells for LFP, NCA, and NCM, where the dashed lines indicate the average RMSE for each model. These results show that including temperature (TEMP) as an input feature allows the model to predict cycle life more accurately and overall error reduction.

rate, may also have a significant influence on features and model performance (Figure S6). Occasionally, the inclusion of temperature did not improve predictions, implying that additional cycling conditions may have a more dominant impact in certain cases. These observations emphasize the importance of considering cycling conditions in machine learning models to ensure their robustness and applicability across different battery materials.

These results indicate that data diversity is the key factor enabling strong performance and cross-material generalization. Incorporating structurally distinct cathode chemistries together with temperature conditions embeds physically meaningful degradation behaviors into the dataset, enabling strong predictive performance without relying on advanced model architectures. In this context, our framework achieves performance on the common LFP dataset (RMSE 102, MAPE 11%)

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that is comparable to state-of-the-art deep learning methods reported in recent literature. For example, BatLiNet [51] reports an averaged error of RMSE 112.5 and MAPE 8.5% on the common LFP dataset. This highlights that a data-centric strategy, using statistical features enriched with material and temperature diversity, can be a powerful and effective path toward improved robustness in battery lifetime prediction.

4. Conclusion

This study provides key insights into developing models that balance accuracy and robustness, providing a foundation for improved battery management strategies. Previous approaches to battery performance prediction have often been limited to specific cathode materials and cycling conditions, which restricted the generalizability of the models. To address this, we adopted a data-centric approach that incorporates diverse cathode materials and cycling temperatures. Our results demonstrate that increasing data diversity reduces both RMSE and MAPE, thereby enhancing the predictive power of the feature. This approach enables the model to capture unique degradation patterns across different materials more accurately. Furthermore, we showed that including cycling temperature as an input feature significantly improved predictive accuracy, emphasizing the importance of accounting for cycling conditions in predictive models. This framework provides interpretability by enabling direct insight into the physical factors that affect prediction performance. Despite these advancements, challenges remain in fully capturing the complex degradation mechanisms across all cathode materials. Future studies could explore advanced features to better represent degradation behaviors and leverage larger, more diverse datasets. Moreover, integrating additional environmental factors beyond just temperature could further enhance model robustness and predictive accuracy. Finally, validating this data-centric approach on standardized open benchmarks, such as BatteryML [52], represents an important next step for confirming its broader generalizability. We hope our findings contribute to the development of machine learning models for battery systems that offer high accuracy, robustness and clear interpretability. These findings lay a foundation for developing robust models adaptable to diverse battery materials and cycling conditions, which are crucial for advancing safer and more efficient battery technology in real-world applications.

Data Availability Statement

Data for this article, including battery cycle life data and degradation features, are available at *Nature Communications* at [https://doi.org/10.1038/s41467-022-29837-w] and *Nature Energy* at [https://doi.org/10.1038/s41560-019-0356-8]. The performance evaluation code used in this study is available at:https://github.com/gaheunn/Data-Driven-Battery-Life-Prediction-Across-Electrode-Materials. The additional data files supporting the findings of this article are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Gaheun Shin: conceptualization, methodology, data curation, writing – original draft. **Joonhee Kang**: conceptualization, writing – review and editing, formal analysis, validation, supervision.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. (*Supporting Information*) Figures S1 and S2 present the distribution of cycle life and the variance of ΔQ_{100-10} (V) across different cathode materials, highlighting their distinct degradation behaviors. Figures S3 to S5 further examine the limitations of models trained on data from a single cathode type and demonstrate how model performance improves as the training dataset is gradually expanded to include multiple materials. Figure S6 shows ΔQ_{100-10} (V) curves under different charging rates, indicating potential rate-dependent degradation patterns. Tables S1 and S2 summarize the correlation analysis between input features and cycle life, as well as the hyperparameters used in machine learning models evaluated in this study.

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